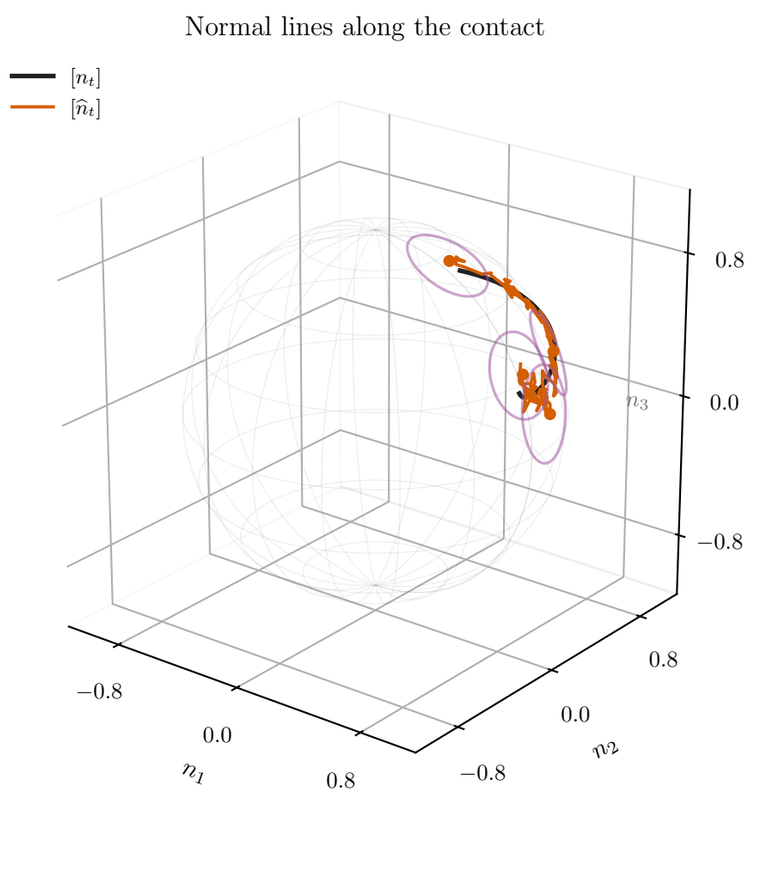
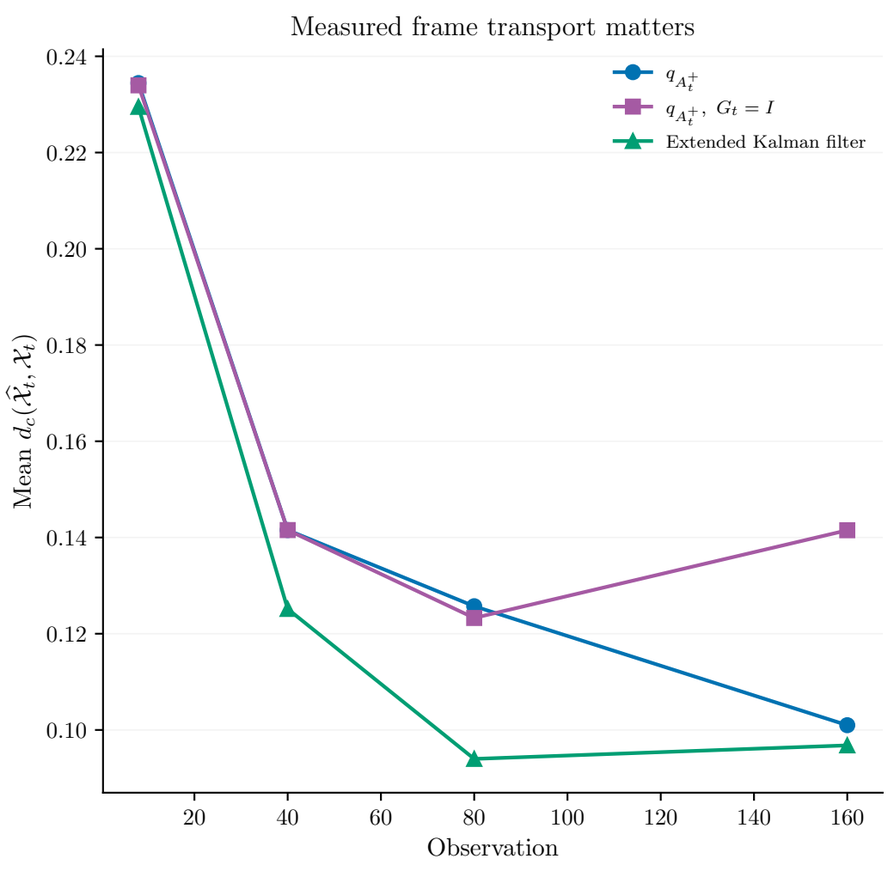
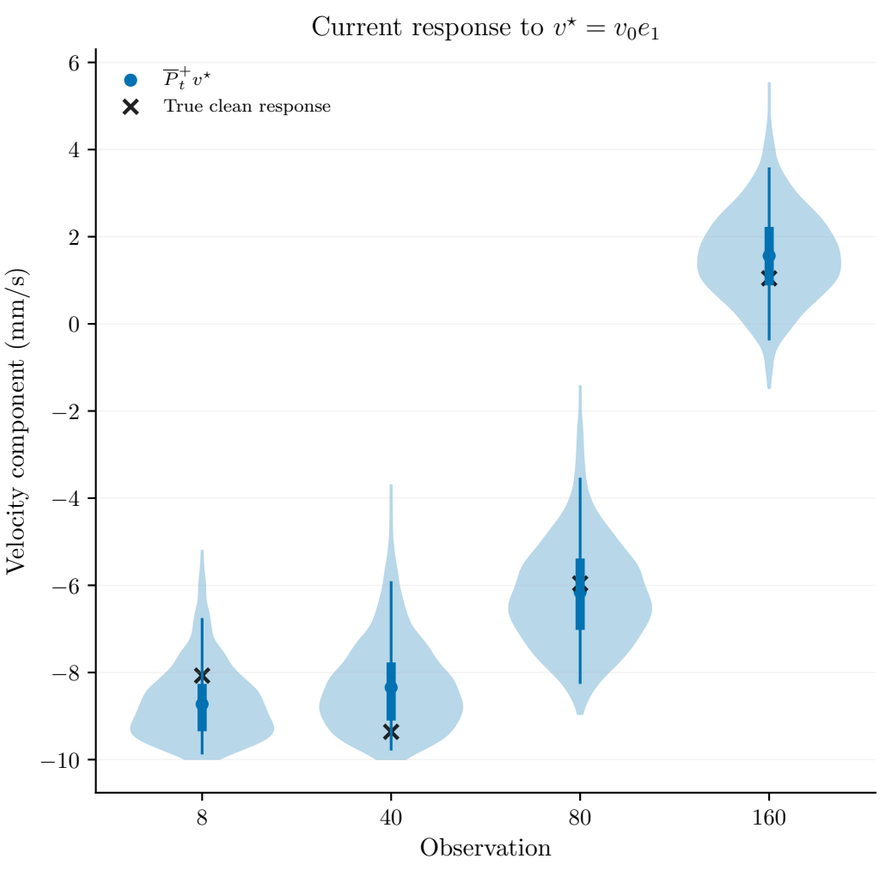

Normal lines along the contact
$[n_t]$
$[\widehat{n}_t]$
0.8
0.0
−0.8
n_3
−0.8
0.0
0.8
n_1
0.8
0.0
−0.8
n_2


Measured frame transport matters
$q_{A_t^+}$
$q_{A_t^+},\ G_t = I$
Extended Kalman filter
Mean $d_c(\widehat{\mathcal{X}}_t, \mathcal{X}_t)$
0.24
0.22
0.20
0.18
0.16
0.14
0.12
0.10
20
40
60
80
100
120
140
160
Observation


Current response to $v^\star = v_0 e_1$
$\overline{P}_t^+ v^\star$
True clean response
Velocity component (mm/s)
6
4
2
0
−2
−4
−6
−8
−10
8
40
80
160
Observation